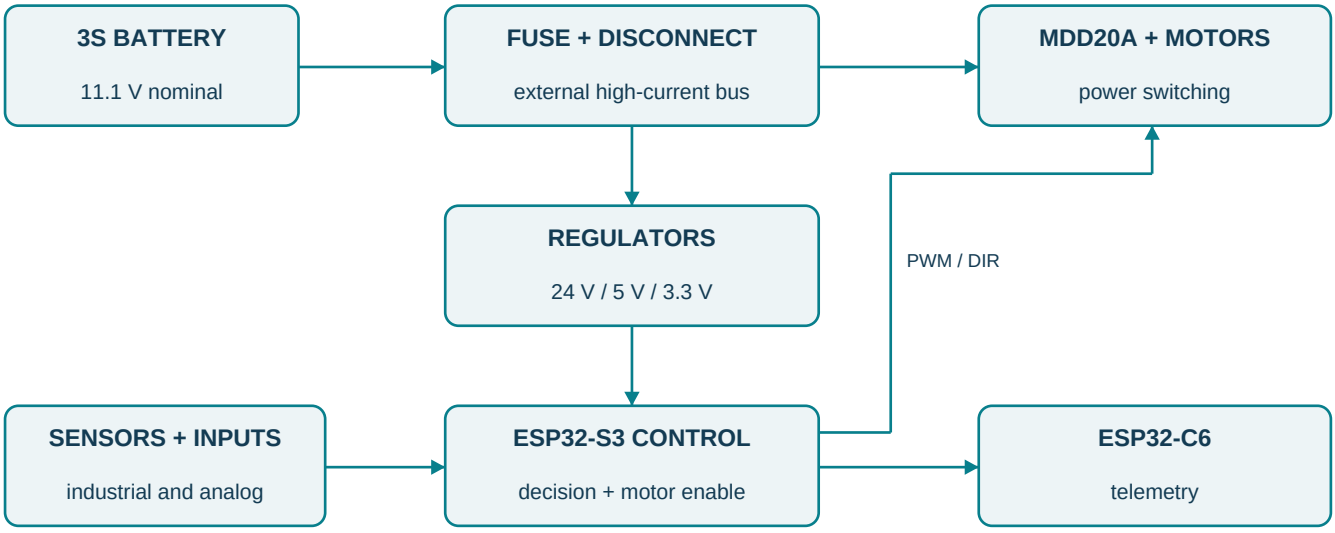
LEVI

From system architecture to every connection

A guided study of the 3 kg robowrestling schematic



Includes worked circuit calculations, a battery recommendation with mechanical data, a component and pin atlas, exercises, and all ten original-size schematic sheets.

Schematic revision 1.0 | Prepared 10 September 2026

Based on the saved KiCad files and exported netlist. Electrical connectivity checked; PCB layout and powered validation remain. Battery recommendation assumes a three-minute match unless your mechanical/runtime requirements say otherwise.

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How to learn Levi

This handbook explains the saved revision 1.0 schematic, not a generic robot circuit. Keep KiCad open beside it. Reference names such as R9 and U30 are search keys: use Find in KiCad to locate them. The reference atlas at the back lists the parts and their actual electrical connections. Full schematic sheets follow the teaching chapters.

Start with three passes. First, spend 20 minutes on the architecture, power tree and battery decision. Second, follow one complete signal from sensor to motor, using the sensor, comparator, ADC and motor chapters. Third, work through the equations, pin atlas and bench exercises. Do not try to memorize 207 PCB components before understanding their repeated patterns.

The schematic is an electrically checked design, not tested hardware. The recorded ERC result is zero errors and warnings; that does not demonstrate immunity to motor noise, adequate thermal margins, successful IO-Link communication or safe fault recovery. This guide marks calculated examples and proposed firmware behavior separately from physical connections already drawn.

Your learning milestones

Stage	You should be able to explain	Practical exercise
System	Where energy flows and who controls motion	Draw the power and information paths from memory
Circuit	Why each component family exists	Trace the left Baumer and left PWM channels
Quantitative	Voltage scaling, filtering, hysteresis and runtime	Recalculate a divider and a three-minute energy budget
Implementation	Pin numbers, startup states and wiring	Find a connector pin and its MCU destination
Validation	What ERC cannot establish	Write a test with an instrument, expected result and stop condition

How to read the pages

Read a circuit in this order: supply voltage, ground return, input source, transformation, output destination, startup state, and fault behavior. Ask what happens when the input is missing, shorted, noisy or late. Each question teaches more than memorizing a component value.

Values such as 10k mean 10,000 ohms; 100n means 100 nanofarads; 1u means 1 microfarad. A voltage rating on a capacitor is a limit, not the voltage it produces. GPIO numbers identify firmware signals; package pin numbers identify physical pads. They are not interchangeable.

The whole robot in two flows

The battery supplies an external MDD20A motor driver and a separately fused electronics board. Three regulators create 24 V for industrial sensing, 5 V for the IO-Link transceivers, and 3.3 V for processors and line sensing. All share an electrical ground reference. Motor current must travel through the heavy external wiring, rather than through the control PCB's power header.

Information flows in the other direction: sensors report the environment, conditioning circuits translate voltage levels, the ESP32-S3 makes a decision, and PWM/direction signals tell the driver how to energize the motors. The second ESP32 handles telemetry. It is not electrically isolated: it shares power and ground with the control MCU.

The nine debug LEDs indicate nine digital logic states. They do not independently prove that a sensor has detected an opponent, that its cable is intact, or that firmware interpreted its polarity correctly.

What each sensor contributes

Device	Quantity	Electrical information available	Route to control MCU
Baumer O300.DI-GM1J.72N	3	Switching output or IO-Link on C/Q	Comparator in DIGITAL mode; MAX14819 in IO-Link mode
Banner Q20NDXL	3	Two complementary NPN outputs per sensor	Six comparator channels
JSumo ML1	4	Analog line/reflectance output	Four divided and filtered ADC1 inputs
Battery divider	1	Total pack voltage	Fifth ADC1 input

The optical sensors require installation and calibration. Their names in the schematic identify electrical channels; the drawing does not define sensor aim, wheel geometry, reflective thresholds or a winning control algorithm. Those depend on the mechanical robot and arena.

A complete example

Suppose the left Baumer's C/Q line becomes HIGH while the selector is in DIGITAL. The signal enters J4 pin 4, reaches the common contact of the left switch pole through J7, and returns as BAUM_L_DIGRAW. R9/R10/C14/R12 condition it; U20A compares it with SENSE_REF. R11 supplies the comparator's HIGH level. BAUM_L reaches S3 GPIO7 and the buffered indicator channel Q1/D6. Firmware may then change PWM_L_RAW and DIR_L_RAW. U40 gates PWM with MOTOR_ENABLE, R69 feeds PWM_L, and J15 connects it to MDD20A PWM1. The driver switches battery current through M1.

Notice the boundaries: the sensor does not directly power the motor, and the MCU does not directly drive motor current. The software decision is a required step between sensing and motion.

Read the schematic like a circuit designer

Symbols, wires and labels

A wire is a conductive connection. A junction dot explicitly joins branches. A crossing without a junction is not an intended connection; use KiCad Highlight Net when uncertain. Labels with the same global name connect electrically even on different sheets. A sheet box on the root page is navigation, not an electronic component.

GND is the common reference node. It is not automatically chassis earth and does not imply that return-current paths have zero impedance. A motor return wire carrying several amps can develop enough voltage to disturb a logic reference. Board layout and harness routing determine that behavior.

An X at a pin is an intentional no-connect marker. Leaving a pin unused is different from tying it LOW. PWR_FLAG is an ERC annotation saying that an external source drives a net; it does not add a regulator, fuse or physical footprint.

Multi-unit components

U20A through U20E are one physical LM339 package: four comparators and a power unit. U30's four units are one MAX14819 package, divided to make the logic, channels and power pins readable. SW101A/B/C are the three mechanically linked poles of one switch. U40's power unit is on sheet 10 while its gates are on sheet 8. Always search all units before deciding that power pins are missing.

Footprints and reference numbering

The symbol describes electrical meaning; the footprint describes copper pads and mounting geometry. A correct pad count alone does not prove that a land pattern has adequate soldering or thermal performance. The local Pololu footprints include the module outline and mounting holes, not just a row of connector pads.

R, C, D, Q, U, J, F, SW, Y and TP identify resistors, capacitors, diodes, transistors, integrated circuits/modules, connectors, fuses, switches, crystals and test points. A1 is the external motor-driver module; BT1 is the battery. Purchased sensors have BOM identifiers but are represented electrically by their board-side harness connectors.

Power: battery to three regulated rails

Start on sheet 9: BT1 positive passes through F100 and the removable power link SW100 to BAT_FUSED. The driver receives this bus directly. J1 brings a small branch to sheet 1. F1 then feeds D1, which feeds LOGIC_IN. D2 and C1 connect from that protected rail to ground.

Part	Actual role	What it does not do
F100, provisional 20 A inline fuse	Limits consequences of a sufficiently large harness fault	Does not detect a motor stalled at 5.5 A
SW100, removable XT60 link	Physically interrupts the positive battery path	Does not discharge every capacitor instantly
F1, 0451002.MRL 2 A	Separate electronics-branch protection	Does not guarantee every regulator's current stays below 2 A
D1, SS34	Blocks reverse input into electronics	Does not protect the external MDD20A against reverse polarity
D2, SMBJ13A	Shunts positive input transients	Is not a precision 13 V regulator
C1, 47 uF / 35 V	Local input energy reservoir	Does not replace a battery or absorb unlimited braking energy
F2, 1812L050/30	Resettable 0.5 A-hold sensor-rail protection	Does not provide instant per-port current limiting

Fuses respond to current and time. A 2 A marking is not a perfect 2.000 A ceiling. A resettable fuse increases resistance as it heats; ambient temperature changes its behavior. Select and test fuse coordination using actual inrush, fault current and wiring.

Why three regulators?

U1, Pololu U3V50F24, boosts the battery-derived input to +24V_RAW. F2 feeds +24V_SENS. U2, Pololu S13V30F5, supplies +5V. U3, Traco TSR2-2433, supplies +3V3. The EN pins of the Pololu modules are left open to use their documented internal enable behavior. These are complete switching-regulator modules; their inductors and switching ICs are inside the modules, not missing PCB parts.

C2 and C5 are 22 uF / 50 V reservoirs for the 24 V rails; C3 is 22 uF / 25 V on 5 V; C4 is 22 uF / 10 V on 3.3 V. Voltage rating alone does not specify capacitance under bias, ESR, temperature performance or ripple-current capability. The current BOM specifies values and packages for many passives, rather than a purchasing-qualified MPN for every capacitor.

Power is conserved, current is not

For a regulator, input current is approximately output power divided by input voltage and efficiency. If the 24 V rail supplies 0.20 A, it delivers 4.8 W. At 10.5 V input and an assumed 85% efficiency, input current is about 0.54 A, before other loads. This is an example, not a measured Levi current budget.

An LDO making 3.3 V from 12 V at 0.5 A would dissipate about 4.35 W. That explains the choice of a switching 3.3 V supply. A 2 A regulator rating is available capacity, not a claim that the circuit always draws 2 A.

Main controller: why the S3 and how it starts

U10 is ESP32-S3-WROOM-1-N8 without PSRAM. The application consumes 28 GPIO signals: five analog inputs, nine digital sensor inputs, seven IO-Link interface signals, five motor signals and two telemetry signals. UART0 programming and BOOT are additional. The choice fits this direct-I/O architecture; it is not a claim that S3 is universally superior to every MCU.

GPIO1, 2, 4, 5 and 6 are used as ADC1 inputs. GPIO35-37 are used for IO-Link; substituting a module that reserves them for octal PSRAM would break the design. GPIO3/45/46 are intentionally unused strapping pins. GPIO0 is BOOT. USB pins 19/20 are unused in this revision.

Reset and boot are different operations

R1 pulls CTRL_EN toward 3.3 V and C9 slows its rise. Their nominal time constant is 10k x 1uF = 10 ms; the actual release time depends on input thresholds and supply ramp. SW1 pulls EN LOW to reset. R2 normally holds GPIO0 HIGH; SW2 pulls BOOT LOW. Holding BOOT while resetting selects the serial download path. This is not an automatic USB programming circuit.

C6, C7 and C8 are 100 nF, 10 uF and 47 uF in parallel across the local supply. The small capacitor helps at higher frequencies when placed close to the module; larger values support slower load changes. Physical trace inductance prevents a distant capacitor from behaving like an ideal capacitor at the pin.

J2 is the control programming header. Its 3.3 V pin is VREF for a compatible adapter, not permission to connect an independent 3.3 V power source. MCU TX connects to adapter RX and vice versa. Use common ground and 3.3 V logic.

Firmware obligations

At startup, keep MOTOR_ARM and both PWM commands LOW. Configure ADC attenuation/calibration, sensor GPIOs, SPI chip selects and timeouts before arming. Hardware PWM generates regular pulses; it does not decide whether movement is appropriate. A proposed control loop should give edge detection priority, reject stale sensor data, handle battery limits and refresh a watchdog. Those behaviors are requirements to implement, not firmware already supplied with this schematic.

Dual-core capability is useful scheduling headroom, but assigning a task to a core does not by itself make it real-time. Interrupt load, locks, flash operations and blocking logging can still introduce latency.

Telemetry and service connections

U11 is ESP32-C6-MINI-1-N4. GPIO18 receives TELEM_RX from S3 GPIO42 through R3. GPIO19 transmits TELEM_TX through R4 to S3 GPIO47. Both series resistors are 100 ohms. They reduce edge-related ringing and limit small contention currents; they are not level shifters or isolation barriers.

R5/C13/SW3 provide reset, R6/SW4 provide BOOT, and R7 holds GPIO8 HIGH for the intended download strapping condition. J3 is a separate UART0 service header. C10-C12 provide local bypassing. A radio-current transient on this module can still affect the shared 3.3 V rail if the supply and layout are poor.

Suggested packet contract

Use a framed message containing a version, sequence number, timestamp, payload length, payload and CRC. Report raw sensor values, interpreted states, battery voltage, commanded PWM, mode and faults. Include a receiver timeout. This is a suggested interface, not an existing software protocol. Losing telemetry must not prevent local edge avoidance.

Connection	Control end	Telemetry end	Meaning
Outbound telemetry	GPIO42 / CTRL_TX	GPIO18 / TELEM_RX	S3 transmits robot state
Return serial data	GPIO47 / CTRL_RX	GPIO19 / TELEM_TX	C6 transmits to S3
Programming	J2 UART0	J3 UART0	Independent firmware loading
Reference	+3V3 and GND	+3V3 and GND	Shared supply, no galvanic isolation

Do not let a debug command bypass the motor-arming rules. A communications processor is a separate scheduling domain, not automatically a safety boundary.

Baumer switching inputs and mode selection

Each Baumer cable uses brown for positive supply, blue for ground, black for C/Q, and white for TEACH. J4, J5 and J6 are board-side four-pin connectors. Their numbered arrangement follows the documented M8 electrical assignments; the PCB connector itself is JST-XH, not an M8 connector.

R8/R13/R18 hold TEACH LOW. TP1/TP2/TP3 expose those lines. They are industrial-voltage signals, not 3.3 V GPIO test points. Do not inject arbitrary teach pulses without following the sensor's teach procedure.

D3-D5 are SMCJ48A TVS devices on the three C/Q lines. Their job is transient suppression. Their value does not mean that normal C/Q is 48 V. ADI discusses this device family for specified surge environments; the complete installed harness still needs a protection and signal-integrity check.

Follow the selector carefully

J7 has three groups: pins 1/2/3 for left, 4/5/6 for center, and 7/8/9 for right. Each group is digital path / sensor C/Q common / IO-Link path. SW101 is one 3PDT switch. The numbers in its custom harness symbol are logical wiring identifiers; identify the physical switch lugs by continuity before assembling the harness.

In DIGITAL, C/Q reaches the comparator input. In IO-Link, it reaches the corresponding MAX14819 C/Q pin. The disconnected digital divider tends toward LOW; it is not a valid sensor reading in that mode. There is no dedicated switch-position input in this schematic, so firmware needs an explicit configuration or a carefully designed mode-detection procedure. Do not assume the MCU can directly read SW101's position.

What the digital conditioner does

The left channel uses R9 = 30k from the raw line, R10 = 10k to ground, C14 = 1 nF to ground, R12 = 1M positive feedback, U20A, and R11 = 4.7k output pull-up. Center and right repeat the circuit with different reference numbers. The next chapter derives it once so you can understand all nine instances.

Comparator circuit: derive every important number

The LM339B compares its positive and negative inputs. Its output is open collector: it can pull LOW, but it cannot actively drive HIGH. R11 supplies the HIGH level from 3.3 V. Powering the comparator from 24 V does not make its pulled-up output 24 V. This is the voltage-domain translation in the design.

R23 and R24 generate the shared reference; C18 filters it. Nominally Vref = 3.3 x 10 / (16.5 + 10) = 1.2453 V. This is a resistor-derived reference, not a precision reference IC. Its voltage follows the 3.3 V rail and resistor tolerances.

Ignoring feedback first, 30k/10k scales a raw input by 1/4. A 24 V input becomes approximately 6 V at the comparator input, which is why the comparator itself is powered from the industrial rail. This node must never be routed directly into the ESP32.

Positive feedback creates hysteresis

At the transition, the positive input equals Vref. Kirchhoff's current law gives:

Vraw,trip = Vref x (1 + 30k/10k + 30k/1M) - Vout x (30k/1M).

With the output LOW at an illustrative 0.2 V, the rising threshold is about 5.012 V. With the output HIGH at an ideal 3.3 V, the falling threshold is about 4.920 V. The separation is about 93 mV referred to the raw input. The actual HIGH is slightly loaded by feedback and the indicator transistor circuit, and actual offsets/tolerances shift these values. These are nominal calculations, not guaranteed limits.

Without hysteresis, a slowly moving or noisy input near threshold could make the output chatter. Positive feedback makes the switching threshold depend on the previous state. The capacitor adds a different effect: it attenuates fast changes continuously.

Filter and loading

With an ideal low-impedance raw source, the resistance seen by C14 is roughly 30k || 10k || 1M = 7.44k. Its time constant is approximately 7.44 us and its single-pole corner is about 21.4 kHz. The source impedance changes that calculation; this matters particularly for the Banner pull-up circuit.

At 24 V, the simple 40k divider path draws around 0.6 mA. The 30k resistor dissipates about 10.8 mW in the no-feedback approximation. R11 supplies roughly (3.3 - 0.2)/4.7k = 0.66 mA when the output is LOW. These values explain why ordinary small resistors are appropriate here.

Unused comparators and failure signatures

Unused units have their positive input grounded, negative input tied to SENSE_REF, and output marked unconnected. Their power pins still belong to the same physical IC. A missing 3.3 V rail can corrupt both Vref and output levels. A missing 24 V rail can invalidate comparator behavior even if the MCU remains powered. Firmware should not equate a stable digital value with proof that all rails are healthy.

Banner NPN inputs: why the pull-ups matter

J8-J10 bring in the three Q20NDXL sensors. Each has a black and white complementary NPN output. An NPN output sinks current when ON; when OFF it needs an external pull-up to establish a HIGH voltage. The six 10k pull-ups connect to +24V_SENS. Six comparator channels then translate the raw lines to 3.3 V logic.

The raw HIGH is not exactly 24 V. A 10k pull-up feeding an approximately 40k divider produces about 19.2 V before feedback and leakage are considered. With an illustrative 200 uA worst-case leakage allowance, that becomes roughly 17.6 V. Both remain well above the approximately 5 V decision threshold. Check the exact output specifications for the purchased suffix. When ON, the sensor pulls the line close to ground and the comparator output becomes LOW.

The 10k pull-up dissipates at most about 24.96 squared / 10,000 = 62 mW if its output is pulled to ground at the upper 24 V regulator tolerance. That is below a nominal 125 mW resistor rating at its specified temperature, although board temperature still requires derating.

Complementary does not mean self-validating

The two outputs should usually represent opposite optical states once switching has settled. Sampling during transitions, disconnecting the sensor, or losing sensor power can produce combinations that are not meaningful detection states. Debounce or validate the pair over time; use a fault state rather than forcing every combination into opponent/no-opponent.

Use the sensor's light/dark operating definition to map electrical HIGH/LOW to detection. An LED marked BAN_L_BK lights for logic HIGH, not automatically for 'opponent left'. Distinguish electrical polarity from semantic meaning in both code and telemetry.

IO-Link: physical layer, framer and software

U30 connects left and center; U31 connects right and has one unused channel. SPI is shared; chip selects and interrupts are separate. The S3 never sees 24 V C/Q directly. The MAX14819 translates between the industrial line and its low-voltage digital interface.

The IC includes framing assistance, but it is not a complete application that automatically returns distance readings when power is applied. The host must configure communication, handle startup and timeouts, process messages, and interpret sensor-specific process data using the correct device documentation/IODD. IO-Link is not interchangeable with a plain 3.3 V UART cable.

Read the four units of one IC

The logic unit contains CS, SCLK, SDI, SDO, IRQ, address and clock pins. Channel units contain C/Q, direct TX/RX controls and optional supply-controller signals. The power unit contains VCC, VL, V5, REGEN, exposed ground and NC pins. The reference atlas lists the exact package pin mapping.

The schematic ties TX/TXEN LOW for the intended internal-framer use, uses separate CS lines with A0/A1 LOW, and leaves unused status outputs disconnected. L+ controllers are omitted: sensors receive shared fused power directly. SN1/SN2 pairs are tied together and otherwise floating. Firmware must disable the unused L+ fault reporting; there is no individual software-switched sensor power channel in this revision.

Supplies and clocks

VCC is 24 V, VL is 3.3 V, and external V5 is 5 V with REGEN grounded. This avoids dissipating the full 24-to-5 V drop in the chip's internal regulator. Local capacitors are distinct from the large supply reservoirs and must be placed by the relevant pins.

Y1/Y2 are specified as 14.7456 MHz crystals with 18 pF load capacitance. C27/C28 and C33/C34 are initial 22 pF external values. For a symmetric oscillator, effective load is approximately half the per-leg capacitance, including internal and parasitic contributions. The design's initial estimate is (22 + 10 + 4)/2 = 18 pF. This is a layout-dependent estimate, not a measured oscillator result. Confirm the ordered crystal package, ESR, startup and frequency.

Why 24 V replaced 12 V

Both industrial sensor models accept the higher rail. With a 12 V supply, the Baumer output drop can leave too little guaranteed HIGH-level margin for the MAX receiver. Raising the rail creates margin; it does not remove the need to check polarity, wiring and surge conditions. The MAX footprint's exposed pad is 5.50 mm for the selected package and must connect to a suitable ground/thermal structure.

Start SPI at the documented design target of 1 MHz, verify register communication, then bring up one IO-Link channel before all three. Exact register values and sensor process-data decoding belong in a separately tested driver; this guide deliberately does not invent unverified register writes.

Analog edge sensing and battery measurement

The four ML1 sensors are supplied from 3.3 V through J11-J14. Their output passes through 10k, then reaches a node with 20k and 10 nF to ground. That node feeds ADC1. Local 100 nF capacitors bypass each sensor supply connection.

Vadc = Vsensor x 20/(10+20) = 2/3 x Vsensor. Therefore a 3.3 V output becomes 2.2 V. The source resistance seen by the capacitor is nominally 10k || 20k = 6.67k, giving tau = 66.7 us and fc = 2.39 kHz. Approximately five time constants, 0.33 ms, brings a step close to its final value when the source behaves ideally.

Do not assume that a 12-bit ADC code directly equals code/4095 x 3.3 V. Attenuation, calibration, nonlinearity and sampling behavior matter. Convert to calibrated millivolts first. Then Vsensor is approximately 1.5 x Vadc. Calibrate thresholds over actual dark/light surfaces at the installed height and ambient lighting.

Battery divider

R67 = 100k, R68 = 22k and C47 = 100 nF. Vadc = Vpack x 22/122. The reverse conversion is Vpack = Vadc x 122/22, or about 5.545 x Vadc. At 12.6 V, Vadc is 2.272 V. At 10.5 V, it is 1.893 V.

The divider continuously draws about 103 uA at 12.6 V while BAT_FUSED is energized. Its nominal filter resistance is 18.03k, time constant 1.80 ms, and corner 88.3 Hz. This helps reject short noise spikes but also delays a sudden voltage change.

Measured ADC voltage	Inferred pack voltage	Interpretation
2.272 V	12.60 V	Nominal full charge of a standard 3S LiPo
2.002 V	11.10 V	Nominal pack voltage, not a precise state of charge
1.893 V	10.50 V	3.5 V/cell average; candidate stop threshold

This circuit measures the sum of all three cells. It cannot identify a weak cell or perform balancing. A pack could have an acceptable total while one cell is too low. Check individual cells with an appropriate balance charger/checker between runs. Select stop thresholds using cell guidance and measured load sag, rather than running down to an absolute minimum.

Motors, PWM and the enable chain

U40 is a dual AND gate. One input receives PWM_L_RAW or PWM_R_RAW; the other receives MOTOR_ENABLE. Each output reaches the driver through a 100-ohm resistor and has a 10k pull-down. DIR_L_RAW and DIR_R_RAW reach the driver through separate 100-ohm resistors.

J16 is a removable link between MOTOR_ARM and MOTOR_ENABLE. R75 and R76 pull those signals LOW. R77-R80 pull the raw PWM and direction commands LOW during reset or high-impedance states. These are passive startup defaults, not firmware supervision.

MOTOR_ENABLE	PWM_RAW	Gated PWM	Result at driver
LOW	LOW or HIGH	LOW	Dynamic brake mode
HIGH	LOW	LOW	Dynamic brake portion of PWM
HIGH	HIGH	HIGH	Drive in the direction selected by DIR

MDD20A PWM LOW means brake, not coast. Therefore removing ARM does not electrically disconnect the motor driver. SW100 is the physical battery disconnect. The driver also has local test buttons; these belong to the external module and must be considered when working near the robot.

PWM is not a speed sensor

Duty cycle controls the driver command. Motor speed still depends on voltage, load, friction and traction. The selected Pololu 4743 has no encoder, and the schematic contains no motor-current sensor. You cannot claim closed-loop wheel speed or measured stall detection from this hardware alone.

At 12 V the motor's published no-load speed is 200 RPM and extrapolated stall current is 5.5 A. A full 3S pack is 12.6 V, slightly above the nominal motor rating. A simple first-order voltage compensation would limit full-pack duty to 12/12.6 = 95.2%, but PWM still applies full bus-voltage pulses; this is not a guarantee of identical motor stress.

Rapid reversal while a motor is spinning can cause a current transient above stationary stall current because back EMF initially opposes the new command. Ramp or manage direction changes, and validate current with instrumentation. The 20 A/channel driver rating is capacity, not what each motor always draws, and its own current limit is not tailored to a 5.5 A-stall motor.

The missing protection boundary

There is no independent external watchdog or per-motor current sensing in revision 1.0. A frozen MCU that leaves MOTOR_ARM HIGH can leave the gate enabled. MCU watchdog reset helps only if correctly configured and able to reset the relevant outputs. For a competition-ready implementation, evaluate independent timeout/current protection against the event's requirements and measured fault behavior. Do not use a prolonged stall as a battery-runtime test.

Debug LEDs, layout and component details

Each digital indicator uses a 47k base resistor, MMBT3904 transistor, green LED and 1k series resistor. The MCU/sensor logic drives the transistor base; the LED current comes from 3.3 V. This prevents the LED itself from becoming a large load on the comparator output.

With an illustrative 0.7 V base-emitter drop, base current is about (3.3 - 0.7)/47k = 55 uA. With an illustrative green LED drop of 2.0 V and transistor drop of 0.2 V, LED current is roughly 1.1 mA. Actual LED color chemistry and transistor gain affect brightness and saturation. LED lit means the electrical net is HIGH.

The transistor symbol pins matter: the atlas gives the actual numbered connections. Do not assume every SOT-23 transistor has the same pin order. Likewise, a diode's cathode marking and an electrolytic capacitor's polarity must match the assembled footprint and rail.

Physical placement changes circuit behavior

Put bypass capacitors near the pins they serve with short ground returns. Route crystal traces short and quiet. Keep antenna areas clear according to the exact module documentation, including nearby metal. Keep sensor return wiring and logic references away from the main motor-current loop. Twisting motor pairs and routing signal/return pairs together can reduce loop area; the final EMI performance must still be measured.

The +24 V boost module is about 48.3 x 15.2 mm in its manufacturer drawing and is taller than a bare PCB. Its footprint outline is not the entire clearance envelope once headers and standoffs are installed. Maintain clearance for mounting hardware and access to connectors.

The SMCJ/SMBJ designations specify package families and electrical families, but final ordering requires a vendor and exact suffix. The BOM's LM339BIDR should be checked against the TI LM339B datasheet; some inherited stock-symbol datasheet fields point to an older generic LM339 document. The guide's source list identifies the intended vendor documentation.

Transient protection is conditional

A TVS has a working standoff voltage, a breakdown region and a higher clamping voltage at a stated pulse current. SMBJ13A's nominal maximum clamp specification is close to the 5 V Pololu module's 22 V input limit. Wiring inductance and actual surge amplitude still need verification. Do not interpret a passing ERC as proof of adequate transient margin. The 24 V supply is also not a certified industrial IO-Link port merely because the nominal voltage is correct.

Battery selection: a concrete starting pack

Recommended baseline: Tattu 1300 mAh 3S 75C with XT60, manufacturer part TA-75C-1300-3S1P-XT60, store SKU TAA13003S75X6. It is a standard 3S LiPo: 11.1 V nominal and 12.6 V fully charged. Published body dimensions are 72 x 36 x 22 mm; mass is 122 g with a stated +/-20 g tolerance. It has XT60 discharge and JST-XHR balance connections. Source B1 is the manufacturer purchase/specification page.

This recommendation assumes approximately a three-minute match, swaps between runs, and space for the pack and protected wiring. Final fit is not established without the battery compartment and remaining robot weight budget. It is not necessary to choose an enormous pack just because the driver says 20 A per channel.

Derive the electrical requirement

At 12.6 V, scaling the published 12 V stationary stall current gives 5.5 x 12.6/12 = 5.78 A per motor, or 11.55 A for two. Allocate an illustrative 2 A battery-side allowance for electronics and conversion losses to obtain a 13.55 A sizing case. This allowance is a conservative scenario, not a measured load or a statement that F1 carries exactly 2 A continuously.

For a three-minute run, required usable charge at 13.55 A is 13.55 x 3/60 = 0.678 Ah. If only 80% of nominal capacity is budgeted, required nominal capacity is about 0.847 Ah. A 1.3 Ah pack provides useful reserve. The motors cannot safely sustain that stall operating point for three minutes; it is a conservative capacity calculation only.

A practical provisional battery target is at least 20 A continuous capability with transient headroom, subject to measurement of reversals, wiring and pack sag. The advertised 75C multiplied by 1.3 Ah is 97.5 A. Treat that as a manufacturer rating, not an independently verified current capability and not the robot's expected draw.

Runtime sensitivity, not a runtime promise

Using 80% of a 1.3 Ah pack, estimated runtime is 60 x 1.3 x 0.8 / average battery amps.

Average battery current	Calculated duration	How to use it
3 A	20.8 minutes	Light-load scenario only
6 A	10.4 minutes	Illustrative mixed driving
10 A	6.24 minutes	Heavy average electrical load
13.55 A	4.61 minutes	Capacity envelope, not permissible continuous stall

Capacity varies with temperature, age, current and voltage cutoff. Real runtime must be measured in a representative drive cycle. Keep setup time and retries inside the energy budget.

Battery alternatives and mechanical integration

All three candidates below are standard 3S, 11.1 V nominal, 75C-labelled packs with XT60. These are alternatives for different space/weight/runtime priorities, not interchangeable mechanical parts.

Candidate	Exact identifier	Published body / mass	Reason to choose
Baseline 1300 mAh	TA-75C-1300-3S1P-XT60 / TAA13003S75X6	72 x 36 x 22 mm; 122 g +/-20 g	Compact starting choice with reserve
Lighter 1050 mAh	TA-75C-1050-3S1P-XT60 / TAA10503S75X6	71 x 36 x 19 mm; 98 g nominal	Save about 24 g; less reserve
Longer 1800 mAh	TA-75C-1800-3S1P-XT60 / TAA18003S75X6	Description: 105 x 35 x 19.3 mm; 159 g +/-20 g	More capacity in a longer, flatter shape

The 1800 mAh page is internally inconsistent: its Specs table says 106 x 36 x 19 mm, with length +/-5 mm and width/height +/-2 mm. Use a conservative body envelope of 111 x 38 x 21.3 mm pending supplier confirmation. Do not silently use the smaller numbers to design a tight pocket.

For the baseline pack, the stated dimensional tolerances give a maximum listed body envelope of 77 x 38 x 24 mm. An initial CAD cavity of about 81 x 42 x 28 mm leaves 2 mm per side beyond that envelope for padding/assembly; that cavity is an engineering starting allowance, not a manufacturer dimension. It does not include the lead exit, XT60, balance plug or bend radius. Add a separate cable/connector bay based on the actual pack. The published discharge lead length is 100 mm with 12 AWG wire; the balance lead is listed as 45 mm.

Use a supported tray, a restraint that prevents movement, smooth insulated surfaces and clearance from screws, gears and sharp edges. Do not clamp a soft LiPo pack rigidly between metal plates or rely on crushing foam to accommodate a pack that no longer fits. Remove damaged or swollen packs from service using manufacturer guidance. Put the mass low and close to the intended center of gravity while protecting it from impact.

The cited manufacturer pages provide dimensions, weight, connector and wire data. They do not establish a certified STEP model or production tolerance drawing. Measure the delivered pack before finalizing its mount. At a stated upper baseline mass of 142 g, plus restraint and wiring, allocate more than the nominal 122 g in the 3 kg mass budget.

Charging and cutoff

Use standard LiPo 3S balance mode, 4.2 V per cell maximum, not LiHV mode. A conservative 1C charge rate for the baseline is 1.3 A if permitted by the delivered pack instructions. Connect the main and balance leads as required by the charger. Charge attended in an appropriate location and store at the manufacturer's storage setting rather than full charge.

The manufacturer manual recommends 3.5 V/cell cutoff for Tattu UAV packs. That corresponds to 10.5 V only when the cells are balanced. Use it as a conservative starting design target, then validate sag filtering and individual-cell behavior; do not treat a total-pack ADC as a balancing or cell-protection system. The MDD20A undervoltage threshold is not a suitable 3S LiPo state-of-charge policy by itself.

Bring-up: learn by measuring

Work from low energy to high energy. This sequence is a proposed validation plan, not a record of tests already performed. Record instrument readings, firmware version and schematic revision so results remain traceable.

Step	Instrument / setup	Expected evidence	Stop and investigate if
Unpowered checks	DMM continuity and inspection	Correct polarities, no hard rail shorts, correct selector mapping	Reversed diode/capacitor, unknown switch lugs or connector mismatch
Electronics rails	Current-limited supply, motors disconnected	About 24 V, 5 V and 3.3 V; sensible idle current	Heating, collapsing rail or current-limit cycling
MCU service	3.3 V UART adapter	Both MCUs can independently reset and load firmware	Boot depends on unplugging unrelated sensors
Analog injection	Known low-voltage source and DMM	ML1 ADC node about 2/3 of injected voltage	Any ADC pin approaches its input limit
Comparator channel	Controlled raw input, scope/DMM	LOW/HIGH transitions around the calculated threshold	Chatter, rail-dependent errors or unexpected polarity
Sensor installation	Actual arena surface and target	Repeatable raw and interpreted states	Thresholds overlap across intended surfaces
IO-Link	Scope/logic analyzer, one sensor first	Reliable SPI, startup, data and timeout recovery	A channel fault blocks all sensing
Motor logic	Scope on J15, driver initially disconnected	No PWM when ARM link absent; correct duty/direction	Startup pulse or persistent enable after reset
Drivetrain	Secured chassis, low duty, current measurement	Correct wheel direction and controlled stopping	Excess current, mechanical binding or unintended movement
Noise and fault tests	Scope plus recorded telemetry	Stable rails, graceful stale-data and reset handling	Radio/motor bursts reset the control MCU

When probing a switching motor output, do not attach an earth-referenced oscilloscope ground clip to an arbitrary motor terminal. Use a suitable differential/isolated measurement arrangement and understand the instrument's grounding. Probe logic relative to the intended ground reference.

The first measurements should establish actual idle electronics current, peak radio current, motor startup current, battery sag and PWM timing. These turn the battery estimate and fuse choice into evidence-based choices.

Troubleshooting by symptom

Symptom	First measurements	Likely paths to investigate
MCU resets when motors start	3.3 V and LOGIC_IN at the MCU/regulator	Ground bounce, wiring resistance, regulator transient response
Baumer always LOW	Selector position, 24 V, raw C/Q, Vref	IO-Link mode selected, cable pinout, teach configuration
Banner polarity seems backwards	Both raw outputs and both GPIO states	Light/dark interpretation versus NPN electrical polarity
Line sensors misclassify edges	Calibrated ADC voltage on real surfaces	Height, ambient light, output range, filter/sampling settings
SPI works but IO-Link does not	Clock, C/Q, framer setup and device identity	Host stack, mode switch, timing, sensor process-data configuration
Motors brake when disabled	PWM LOW on J15	Expected MDD20A behavior; not coast
LED and software disagree	Probe the named GPIO net	Sampling/semantic inversion or wrong channel mapping
Pack voltage looks fine but runtime is poor	Per-cell voltage and loaded sag	Weak cell, ageing, excess mechanical load, wrong current assumption

Use binary isolation: establish the signal at the connector, then after conditioning, then at the MCU, then in software. Replacing components before locating the boundary often hides the original problem.

Exercises and worked answers

Try these without reading the answers. Use the schematic and a calculator, then explain the result aloud.

1. An ML1 produces 2.7 V. What should its ADC node be? What sensor voltage corresponds to a calibrated ADC reading of 1.4 V?

2. BAT_ADC measures 2.0 V. What pack voltage does that imply? Can it prove all cells are healthy?

3. Why does a 24 V-powered comparator deliver a 3.3 V HIGH? What happens if its output pull-up is missing?

4. If the 1M positive-feedback resistor were removed, what changes and what stays the same?

5. Both motors draw 5.5 A. Why might a 20 A main fuse remain intact while the motors overheat?

6. MOTOR_ARM is HIGH but the removable ARM link is absent. What reaches the driver PWM input?

7. How much nominal capacity supports a three-minute, 10 A average run with 20% reserve?

8. In IO-Link mode, can firmware use BAUM_L as a valid digital opponent input? Can it directly read the switch position?

9. How can two processors still fail together despite separate firmware?

10. Why is 75C x 1.3 Ah not a prediction that Levi draws 97.5 A?

Answers

1. 1.8 V at the ADC; 2.1 V inferred sensor output. The divider ratio is 2/3.

2. About 11.09 V. No: it measures total voltage, not individual cells.

3. The open collector pulls down while the resistor pulls up to 3.3 V. Without the pull-up, HIGH is not reliably established.

4. Hysteresis disappears. The divider and capacitor still scale/filter the input; the nominal switching reference remains.

5. The combined 11 A is below 20 A. Fuse time-current behavior and motor thermal limits are different protections.

6. R76 holds MOTOR_ENABLE LOW; the AND output is LOW, commanding the driver's brake state.

7. 10 x 3/60 / 0.8 = 0.625 Ah, before extra operational reserve and ageing allowance.

8. No to both in the current hardware. The digital path is disconnected and no mode-sense GPIO exists.

9. They share the supply, ground, environment and serial connection; electrical faults can cross those boundaries.

10. C-rating is a claimed delivery capability. The load determines current, subject to voltage, resistance and dynamics.

Final teach-back challenge

Explain one left-side sensing-to-motion path, then explain why the robot remains stopped during programming. Point to every component responsible. Finally, identify which part of your explanation depends on firmware and which part depends on physical wiring. If you can do that, you understand the architecture rather than only its labels.

Sources and how to use the reference atlas

The following links identify primary documentation and battery specifications checked on 2026-09-10. Product pages can change; match the exact suffix and delivered label. The project files are the source for the actual connections and component values in this handbook.

ID	Source / purpose
S1	ESP32-S3-WROOM-1 module datasheet: pin allocation, boot and module constraints
S2	ESP32-C6-MINI-1 module datasheet: pin allocation, boot and service interface
S3	Analog Devices MAX14819/MAX14819A datasheet: physical layer, clock, supply and framer requirements
S4	TI LM339B datasheet: comparator behavior and electrical limits
S5	Baumer O300 sensor datasheet: cable assignments and operating requirements
S6	Banner Q20 manual: NPN outputs and wiring
S7	JSumo ML1 product documentation: sensor supply and interface
S8	Cytron MDD20A manual: terminal assignments and brake/drive behavior
S9	Pololu 4743 product page: motor electrical/mechanical characteristics
S10	Pololu U3V50F24 and mechanical drawing: boost module and body geometry
S11	Traco TSR2 datasheet: 3.3 V regulator
B1	Tattu 1300 mAh exact product page: recommended pack, dimensions and tolerances
B2	Tattu 1050 mAh manufacturer page: lighter alternative
B3	Tattu 1800 mAh exact product page: longer alternative and conflicting dimensions
B4	Gens ace/Tattu battery manual: charging and cutoff guidance

The next sections are generated from the saved files. A net named Net-(...) is a local wire without a human-assigned label, not an unresolved connection. NC means explicitly marked unconnected. External harness parts are intentionally excluded from the PCB netlist; their wiring is shown on sheet 9. The complete symbol-pin atlas includes unused and power pins so you can distinguish intentional omissions from missing information.

Primary source links

S1 ESP32-S3 module

S2 ESP32-C6 module

S3 MAX14819

S4 TI LM339B

S5 Baumer O300

S6 Banner Q20

S7 JSumo ML1

S8 Cytron MDD20A

S9 Pololu 4743 motor

S10 Pololu U3V50F24

S10b Boost mechanical drawing

S11 Traco TSR2

B1 Recommended 1300 mAh pack

B2 Lighter 1050 mAh pack

B3 Longer 1800 mAh pack

B4 Battery manual

B5 Manufacturer battery guide

Component reference atlas

All 226 BOM entries are listed, including 207 PCB components and purchased external items. Repeated circuits are explained once in the teaching chapters; this atlas locates each physical part. Pin numbers below are physical symbol/footprint identifiers. The footprint dictionary follows the ledger.

Levi

Ref / value	Purpose	Connections / footprint
C1 / 47u 35V	Supply bypass / local energy reservoir.	1: LOGIC_IN; 2: GND FP4
C2 / 22u 50V	Supply bypass / local energy reservoir.	1: +24V_RAW; 2: GND FP5
C3 / 22u 25V	Supply bypass / local energy reservoir.	1: +5V; 2: GND FP2
C4 / 22u 10V	Supply bypass / local energy reservoir.	1: +3V3; 2: GND FP2
C5 / 22u 50V	Supply bypass / local energy reservoir.	1: +24V_SENS; 2: GND FP5
D1 / SS34	Reverse-input blocking.	1: LOGIC_IN; 2: Net-(D1-A) FP14
D2 / SMBJ13A	Transient suppression; not a precision clamp.	1: LOGIC_IN; 2: GND FP15
F1 / 0451002.MRL / 2A	Branch overcurrent protection; check time-current behavior.	1: Net-(D1-A); 2: BAT_FUSED FP18
F2 / 1812L050/30 / 0.5A	Branch overcurrent protection; check time-current behavior.	1: +24V_RAW; 2: +24V_SENS FP17
J1 / B2B-XH-A | power	Board harness/service interface; exact pin map below.	1: BAT_FUSED; 2: GND FP6
U1 / U3V50F24	24 V boost module.	See full pin atlas. FP22
U2 / S13V30F5	5 V buck-boost module.	See full pin atlas. FP21
U3 / TSR2-2433	3.3 V switching module.	1: LOGIC_IN; 2: GND; 3: +3V3 FP12

01_Control

Ref / value	Purpose	Connections / footprint
C6 / 100n 10V	Supply bypass / local energy reservoir.	1: +3V3; 2: GND FP2
C7 / 10u 10V	Supply bypass / local energy reservoir.	1: +3V3; 2: GND FP2
C8 / 47u 10V	Supply bypass / local energy reservoir.	1: +3V3; 2: GND FP3
C9 / 1u 10V	EN reset RC capacitor.	1: CTRL_EN; 2: GND FP2
J2 / B6B-XH-A | program	Board harness/service interface; exact pin map below.	See full pin atlas. FP10
R1 / 10k	EN pull-up; with C gives reset ramp.	1: +3V3; 2: CTRL_EN FP28
R2 / 10k	BOOT/strap default HIGH.	1: +3V3; 2: CTRL_BOOT FP28
R3 / 100R	Inter-MCU UART series damping.	1: CTRL_TX; 2: TELEM_RX FP28
R4 / 100R	Inter-MCU UART series damping.	1: TELEM_TX; 2: CTRL_RX FP28

Ref / value	Purpose	Connections / footprint
SW1 / RESET	Reset / boot switch.	1: CTRL_EN; 2: GND FP1
SW2 / BOOT	Reset / boot switch.	1: CTRL_BOOT; 2: GND FP1
U10 / ESP32-S3-WROOM-1-N8	Main controller.	See full pin atlas. FP27

02_Telemetry

Ref / value	Purpose	Connections / footprint
C10 / 100n 10V	Supply bypass / local energy reservoir.	1: +3V3; 2: GND FP2
C11 / 10u 10V	Supply bypass / local energy reservoir.	1: +3V3; 2: GND FP2
C12 / 47u 10V	Supply bypass / local energy reservoir.	1: +3V3; 2: GND FP3
C13 / 1u 10V	EN reset RC capacitor.	1: TEL_EN; 2: GND FP2
J3 / B6B-XH-A | program	Board harness/service interface; exact pin map below.	See full pin atlas. FP10
R5 / 10k	EN pull-up; with C gives reset ramp.	1: +3V3; 2: TEL_EN FP28
R6 / 10k	BOOT/strap default HIGH.	1: +3V3; 2: TEL_BOOT FP28
R7 / 10k	BOOT/strap default HIGH.	1: +3V3; 2: TEL_STRAP8 FP28
SW3 / RESET	Reset / boot switch.	1: TEL_EN; 2: GND FP1
SW4 / BOOT	Reset / boot switch.	1: TEL_BOOT; 2: GND FP1
U11 / ESP32-C6-MINI-1-N4	Telemetry processor.	See full pin atlas. FP26

03_Baumer

Ref / value	Purpose	Connections / footprint
C14 / 1n 50V C0G	Raw comparator input noise filter.	1: Net-(U20A-+); 2: GND FP2
C15 / 1n 50V C0G	Raw comparator input noise filter.	1: Net-(U20B-+); 2: GND FP2
C16 / 1n 50V C0G	Raw comparator input noise filter.	1: Net-(U20C-+); 2: GND FP2
C17 / 100n 50V	Supply bypass / local energy reservoir.	1: +24V_SENS; 2: GND FP2
C18 / 100n 10V	Shared comparator reference filtering.	1: SENSE_REF; 2: GND FP2
D3 / SMCJ48A	Transient suppression; not a precision clamp.	1: CQ_L; 2: GND FP16
D4 / SMCJ48A	Transient suppression; not a precision clamp.	1: CQ_C; 2: GND FP16
D5 / SMCJ48A	Transient suppression; not a precision clamp.	1: CQ_R; 2: GND FP16
J4 / B4B-XH-A | L	Board harness/service interface; exact pin map below.	See full pin atlas. FP8
J5 / B4B-XH-A | C	Board harness/service interface; exact pin map below.	See full pin atlas. FP8

Ref / value	Purpose	Connections / footprint
J6 / B4B-XH-A | R	Board harness/service interface; exact pin map below.	See full pin atlas. FP8
J7 / B9B-XH-A | selector	Board harness/service interface; exact pin map below.	See full pin atlas. FP11
R8 / 10k	Baumer TEACH default LOW.	1: TEACH_L; 2: GND FP28
R9 / 30k 1%	Raw-input divider series arm.	1: Net-(U20A-+); 2: BAUM_L_DIGRAW FP28
R10 / 10k 1%	Raw-input divider shunt arm.	1: Net-(U20A-+); 2: GND FP28
R11 / 4.7k	3.3 V comparator output pull-up.	1: +3V3; 2: BAUM_L FP28
R12 / 1M 1%	Positive feedback / hysteresis.	1: BAUM_L; 2: Net-(U20A-+) FP28
R13 / 10k	Baumer TEACH default LOW.	1: TEACH_C; 2: GND FP28
R14 / 30k 1%	Raw-input divider series arm.	1: Net-(U20B-+); 2: BAUM_C_DIGRAW FP28
R15 / 10k 1%	Raw-input divider shunt arm.	1: Net-(U20B-+); 2: GND FP28
R16 / 4.7k	3.3 V comparator output pull-up.	1: +3V3; 2: BAUM_C FP28
R17 / 1M 1%	Positive feedback / hysteresis.	1: BAUM_C; 2: Net-(U20B-+) FP28
R18 / 10k	Baumer TEACH default LOW.	1: TEACH_R; 2: GND FP28
R19 / 30k 1%	Raw-input divider series arm.	1: Net-(U20C-+); 2: BAUM_R_DIGRAW FP28
R20 / 10k 1%	Raw-input divider shunt arm.	1: Net-(U20C-+); 2: GND FP28
R21 / 4.7k	3.3 V comparator output pull-up.	1: +3V3; 2: BAUM_R FP28
R22 / 1M 1%	Positive feedback / hysteresis.	1: BAUM_R; 2: Net-(U20C-+) FP28
R23 / 16.5k 1%	Shared 1.245 V reference divider.	1: +3V3; 2: SENSE_REF FP28
R24 / 10k 1%	Shared 1.245 V reference divider.	1: SENSE_REF; 2: GND FP28
TP1 / TEACH L	Baumer industrial-voltage teach access.	1: TEACH_L FP29
TP2 / TEACH C	Baumer industrial-voltage teach access.	1: TEACH_C FP29
TP3 / TEACH R	Baumer industrial-voltage teach access.	1: TEACH_R FP29
U20 / LM339BIDR	Three Baumer comparators plus unused unit.	See full pin atlas. FP23

04_Banner

Ref / value	Purpose	Connections / footprint
C19 / 1n 50V C0G	Raw comparator input noise filter.	1: Net-(U21A-+); 2: GND FP2
C20 / 1n 50V C0G	Raw comparator input noise filter.	1: Net-(U21B-+); 2: GND FP2
C21 / 1n 50V C0G	Raw comparator input noise filter.	1: Net-(U21C-+); 2: GND FP2
C22 / 1n 50V C0G	Raw comparator input noise filter.	1: Net-(U21D-+); 2: GND FP2
C23 / 1n 50V C0G	Raw comparator input noise filter.	1: Net-(U22A-+); 2: GND FP2
C24 / 1n 50V C0G	Raw comparator input noise filter.	1: Net-(U22B-+); 2: GND FP2

Ref / value	Purpose	Connections / footprint
C25 / 100n 50V	Supply bypass / local energy reservoir.	1: +24V_SENS; 2: GND FP2
C26 / 100n 50V	Supply bypass / local energy reservoir.	1: +24V_SENS; 2: GND FP2
J8 / B4B-XH-A | L	Board harness/service interface; exact pin map below.	See full pin atlas. FP8
J9 / B4B-XH-A | C	Board harness/service interface; exact pin map below.	See full pin atlas. FP8
J10 / B4B-XH-A | R	Board harness/service interface; exact pin map below.	See full pin atlas. FP8
R25 / 30k 1%	Raw-input divider series arm.	1: Net-(U21A-+); 2: BN_L_BK_RAW FP28
R26 / 10k 1%	Raw-input divider shunt arm.	1: Net-(U21A-+); 2: GND FP28
R27 / 4.7k	3.3 V comparator output pull-up.	1: +3V3; 2: BAN_L_BK FP28
R28 / 1M 1%	Positive feedback / hysteresis.	1: BAN_L_BK; 2: Net-(U21A-+) FP28
R29 / 10k	24 V Banner NPN pull-up.	1: +24V_SENS; 2: BN_L_BK_RAW FP28
R30 / 30k 1%	Raw-input divider series arm.	1: Net-(U21B-+); 2: BN_L_WH_RAW FP28
R31 / 10k 1%	Raw-input divider shunt arm.	1: Net-(U21B-+); 2: GND FP28
R32 / 4.7k	3.3 V comparator output pull-up.	1: +3V3; 2: BAN_L_WH FP28
R33 / 1M 1%	Positive feedback / hysteresis.	1: BAN_L_WH; 2: Net-(U21B-+) FP28
R34 / 10k	24 V Banner NPN pull-up.	1: +24V_SENS; 2: BN_L_WH_RAW FP28
R35 / 30k 1%	Raw-input divider series arm.	1: Net-(U21C-+); 2: BN_C_BK_RAW FP28
R36 / 10k 1%	Raw-input divider shunt arm.	1: Net-(U21C-+); 2: GND FP28
R37 / 4.7k	3.3 V comparator output pull-up.	1: +3V3; 2: BAN_C_BK FP28
R38 / 1M 1%	Positive feedback / hysteresis.	1: BAN_C_BK; 2: Net-(U21C-+) FP28
R39 / 10k	24 V Banner NPN pull-up.	1: +24V_SENS; 2: BN_C_BK_RAW FP28
R40 / 30k 1%	Raw-input divider series arm.	1: Net-(U21D-+); 2: BN_C_WH_RAW FP28
R41 / 10k 1%	Raw-input divider shunt arm.	1: Net-(U21D-+); 2: GND FP28
R42 / 4.7k	3.3 V comparator output pull-up.	1: +3V3; 2: BAN_C_WH FP28
R43 / 1M 1%	Positive feedback / hysteresis.	1: BAN_C_WH; 2: Net-(U21D-+) FP28
R44 / 10k	24 V Banner NPN pull-up.	1: +24V_SENS; 2: BN_C_WH_RAW FP28
R45 / 30k 1%	Raw-input divider series arm.	1: Net-(U22A-+); 2: BN_R_BK_RAW FP28
R46 / 10k 1%	Raw-input divider shunt arm.	1: Net-(U22A-+); 2: GND FP28
R47 / 4.7k	3.3 V comparator output pull-up.	1: +3V3; 2: BAN_R_BK FP28
R48 / 1M 1%	Positive feedback / hysteresis.	1: BAN_R_BK; 2: Net-(U22A-+) FP28
R49 / 10k	24 V Banner NPN pull-up.	1: +24V_SENS; 2: BN_R_BK_RAW FP28
R50 / 30k 1%	Raw-input divider series arm.	1: Net-(U22B-+); 2: BN_R_WH_RAW FP28
R51 / 10k 1%	Raw-input divider shunt arm.	1: Net-(U22B-+); 2: GND FP28

Ref / value	Purpose	Connections / footprint
R52 / 4.7k	3.3 V comparator output pull-up.	1: +3V3; 2: BAN_R_WH FP28
R53 / 1M 1%	Positive feedback / hysteresis.	1: BAN_R_WH; 2: Net-(U22B-+) FP28
R54 / 10k	24 V Banner NPN pull-up.	1: +24V_SENS; 2: BN_R_WH_RAW FP28
U21 / LM339BIDR	Four Banner comparators.	See full pin atlas. FP23
U22 / LM339BIDR	Two Banner comparators plus unused units.	See full pin atlas. FP23

05_IOLink_A

Ref / value	Purpose	Connections / footprint
C27 / 22p C0G	Crystal load capacitor.	1: PHY0_XI; 2: GND FP2
C28 / 22p C0G	Crystal load capacitor.	1: PHY0_XO; 2: GND FP2
C29 / 1u 50V	Supply bypass / local energy reservoir.	1: +24V_SENS; 2: GND FP2
C30 / 1u 50V	Supply bypass / local energy reservoir.	1: +24V_SENS; 2: GND FP2
C31 / 1u 10V	Supply bypass / local energy reservoir.	1: +5V; 2: GND FP2
C32 / 100n 10V	Supply bypass / local energy reservoir.	1: +3V3; 2: GND FP2
R55 / 10k	IO-Link CS default HIGH.	1: +3V3; 2: IOL_CS_A FP28
R56 / 4.7k	Open-drain IRQ pull-up.	1: +3V3; 2: IOL_IRQ_A FP28
U30 / MAX14819AATM	Left/center IO-Link master.	See full pin atlas. FP20
Y1 / 14.7456MHz CL18pF	IO-Link reference crystal; load tuning required.	1: PHY0_XI; 2: PHY0_XO FP13

06_IOLink_B

Ref / value	Purpose	Connections / footprint
C33 / 22p C0G	Crystal load capacitor.	1: PHY1_XI; 2: GND FP2
C34 / 22p C0G	Crystal load capacitor.	1: PHY1_XO; 2: GND FP2
C35 / 1u 50V	Supply bypass / local energy reservoir.	1: +24V_SENS; 2: GND FP2
C36 / 1u 50V	Supply bypass / local energy reservoir.	1: +24V_SENS; 2: GND FP2
C37 / 1u 10V	Supply bypass / local energy reservoir.	1: +5V; 2: GND FP2
C38 / 100n 10V	Supply bypass / local energy reservoir.	1: +3V3; 2: GND FP2
R57 / 10k	IO-Link CS default HIGH.	1: +3V3; 2: IOL_CS_B FP28
R58 / 4.7k	Open-drain IRQ pull-up.	1: +3V3; 2: IOL_IRQ_B FP28
U31 / MAX14819AATM	Right IO-Link master; spare channel unused.	See full pin atlas. FP20
Y2 / 14.7456MHz CL18pF	IO-Link reference crystal; load tuning required.	1: PHY1_XI; 2: PHY1_XO FP13

07_Line_Motor

Ref / value	Purpose	Connections / footprint
C39 / 10n 10V	ML1 ADC low-pass capacitor.	1: LINE_FL_ADC; 2: GND FP2
C40 / 100n 10V	Supply bypass / local energy reservoir.	1: +3V3; 2: GND FP2
C41 / 10n 10V	ML1 ADC low-pass capacitor.	1: LINE_FR_ADC; 2: GND FP2
C42 / 100n 10V	Supply bypass / local energy reservoir.	1: +3V3; 2: GND FP2
C43 / 10n 10V	ML1 ADC low-pass capacitor.	1: LINE_RL_ADC; 2: GND FP2
C44 / 100n 10V	Supply bypass / local energy reservoir.	1: +3V3; 2: GND FP2
C45 / 10n 10V	ML1 ADC low-pass capacitor.	1: LINE_RR_ADC; 2: GND FP2
C46 / 100n 10V	Supply bypass / local energy reservoir.	1: +3V3; 2: GND FP2
C47 / 100n 10V	Battery ADC low-pass capacitor.	1: BAT_ADC; 2: GND FP2
J11 / B3B-XH-A | FL	Board harness/service interface; exact pin map below.	1: +3V3; 2: GND; 3: LINE_FL_RAW FP7
J12 / B3B-XH-A | FR	Board harness/service interface; exact pin map below.	1: +3V3; 2: GND; 3: LINE_FR_RAW FP7
J13 / B3B-XH-A | RL	Board harness/service interface; exact pin map below.	1: +3V3; 2: GND; 3: LINE_RL_RAW FP7
J14 / B3B-XH-A | RR	Board harness/service interface; exact pin map below.	1: +3V3; 2: GND; 3: LINE_RR_RAW FP7
J15 / MDD20A control	Board harness/service interface; exact pin map below.	See full pin atlas. FP9
R59 / 10k 1%	ML1 2/3 ADC scaling divider.	1: LINE_FL_ADC; 2: LINE_FL_RAW FP28
R60 / 20k 1%	ML1 2/3 ADC scaling divider.	1: LINE_FL_ADC; 2: GND FP28
R61 / 10k 1%	ML1 2/3 ADC scaling divider.	1: LINE_FR_ADC; 2: LINE_FR_RAW FP28
R62 / 20k 1%	ML1 2/3 ADC scaling divider.	1: LINE_FR_ADC; 2: GND FP28
R63 / 10k 1%	ML1 2/3 ADC scaling divider.	1: LINE_RL_ADC; 2: LINE_RL_RAW FP28
R64 / 20k 1%	ML1 2/3 ADC scaling divider.	1: LINE_RL_ADC; 2: GND FP28
R65 / 10k 1%	ML1 2/3 ADC scaling divider.	1: LINE_RR_ADC; 2: LINE_RR_RAW FP28
R66 / 20k 1%	ML1 2/3 ADC scaling divider.	1: LINE_RR_ADC; 2: GND FP28
R67 / 100k 1%	Battery 22/122 scaling divider.	1: BAT_FUSED; 2: BAT_ADC FP28
R68 / 22k 1%	Battery 22/122 scaling divider.	1: BAT_ADC; 2: GND FP28
R69 / 100R	Motor command series damping.	1: PWM_L; 2: Net-(R69-Pad2) FP28
R70 / 10k	Motor command/enable default LOW.	1: PWM_L; 2: GND FP28
R71 / 100R	Motor command series damping.	1: DIR_L_RAW; 2: DIR_L FP28
R72 / 100R	Motor command series damping.	1: PWM_R; 2: Net-(R72-Pad2) FP28
R73 / 10k	Motor command/enable default LOW.	1: PWM_R; 2: GND FP28
R74 / 100R	Motor command series damping.	1: DIR_R_RAW; 2: DIR_R FP28

08_Harness

Ref / value	Purpose	Connections / footprint
A1 / Cytron_MDD20A	External dual motor power stage.	External; see harness/sensor chapter. No PCB footprint
BT1 / 3S LiPo / XT60	3S battery; handbook recommends Tattu 1300 mAh.	External; see harness/sensor chapter. No PCB footprint
C49 / 1000u 35V low ESR	External motor-bus bulk reservoir.	External; see harness/sensor chapter. No PCB footprint
D15 / SMBJ15A / harness board	Transient suppression; not a precision clamp.	External; see harness/sensor chapter. No PCB footprint
F100 / ATC 20A inline	Branch overcurrent protection; check time-current behavior.	External; see harness/sensor chapter. No PCB footprint
M1 / Pololu 4743 / 50:1	External brushed gearmotor; no encoder in selected part.	External; see harness/sensor chapter. No PCB footprint
M2 / Pololu 4743 / 50:1	External brushed gearmotor; no encoder in selected part.	External; see harness/sensor chapter. No PCB footprint
SW100 / XT60 removable power link	Physical battery disconnect.	External; see harness/sensor chapter. No PCB footprint
SW101 / 50209LX	Mechanically linked 3PDT mode selector.	External; see harness/sensor chapter. No PCB footprint

09_Enable_Debug

Ref / value	Purpose	Connections / footprint
C48 / 100n 10V	Supply bypass / local energy reservoir.	1: +3V3; 2: GND FP2
D6 / GREEN	HIGH-state indicator LED.	1: Net-(D6-K); 2: Net-(D6-A) FP19
D7 / GREEN	HIGH-state indicator LED.	1: Net-(D7-K); 2: Net-(D7-A) FP19
D8 / GREEN	HIGH-state indicator LED.	1: Net-(D8-K); 2: Net-(D8-A) FP19
D9 / GREEN	HIGH-state indicator LED.	1: Net-(D9-K); 2: Net-(D9-A) FP19
D10 / GREEN	HIGH-state indicator LED.	1: Net-(D10-K); 2: Net-(D10-A) FP19
D11 / GREEN	HIGH-state indicator LED.	1: Net-(D11-K); 2: Net-(D11-A) FP19
D12 / GREEN	HIGH-state indicator LED.	1: Net-(D12-K); 2: Net-(D12-A) FP19
D13 / GREEN	HIGH-state indicator LED.	1: Net-(D13-K); 2: Net-(D13-A) FP19
D14 / GREEN	HIGH-state indicator LED.	1: Net-(D14-K); 2: Net-(D14-A) FP19
J16 / ARM removable link	Board harness/service interface; exact pin map below.	1: MOTOR_ARM; 2: MOTOR_ENABLE FP6
Q1 / MMBT3904	Buffered sensor-logic LED driver.	1: Net-(Q1-B); 2: GND; 3: Net-(D6-K) FP25
Q2 / MMBT3904	Buffered sensor-logic LED driver.	1: Net-(Q2-B); 2: GND; 3: Net-(D7-K) FP25
Q3 / MMBT3904	Buffered sensor-logic LED driver.	1: Net-(Q3-B); 2: GND; 3: Net-(D8-K) FP25

Ref / value	Purpose	Connections / footprint
Q4 / MMBT3904	Buffered sensor-logic LED driver.	1: Net-(Q4-B); 2: GND; 3: Net-(D9-K) FP25
Q5 / MMBT3904	Buffered sensor-logic LED driver.	1: Net-(Q5-B); 2: GND; 3: Net-(D10-K) FP25
Q6 / MMBT3904	Buffered sensor-logic LED driver.	1: Net-(Q6-B); 2: GND; 3: Net-(D11-K) FP25
Q7 / MMBT3904	Buffered sensor-logic LED driver.	1: Net-(Q7-B); 2: GND; 3: Net-(D12-K) FP25
Q8 / MMBT3904	Buffered sensor-logic LED driver.	1: Net-(Q8-B); 2: GND; 3: Net-(D13-K) FP25
Q9 / MMBT3904	Buffered sensor-logic LED driver.	1: Net-(Q9-B); 2: GND; 3: Net-(D14-K) FP25
R75 / 10k	Motor command/enable default LOW.	1: MOTOR_ARM; 2: GND FP28
R76 / 10k	Motor command/enable default LOW.	1: MOTOR_ENABLE; 2: GND FP28
R77 / 10k	Motor command/enable default LOW.	1: PWM_L_RAW; 2: GND FP28
R78 / 10k	Motor command/enable default LOW.	1: PWM_R_RAW; 2: GND FP28
R79 / 10k	Motor command/enable default LOW.	1: DIR_L_RAW; 2: GND FP28
R80 / 10k	Motor command/enable default LOW.	1: DIR_R_RAW; 2: GND FP28
R81 / 47k	Indicator base-current limiter.	1: Net-(Q1-B); 2: BAUM_L FP28
R82 / 1k	Indicator LED current limiter.	1: +3V3; 2: Net-(D6-A) FP28
R83 / 47k	Indicator base-current limiter.	1: Net-(Q2-B); 2: BAUM_C FP28
R84 / 1k	Indicator LED current limiter.	1: +3V3; 2: Net-(D7-A) FP28
R85 / 47k	Indicator base-current limiter.	1: Net-(Q3-B); 2: BAUM_R FP28
R86 / 1k	Indicator LED current limiter.	1: +3V3; 2: Net-(D8-A) FP28
R87 / 47k	Indicator base-current limiter.	1: Net-(Q4-B); 2: BAN_L_BK FP28
R88 / 1k	Indicator LED current limiter.	1: +3V3; 2: Net-(D9-A) FP28
R89 / 47k	Indicator base-current limiter.	1: Net-(Q5-B); 2: BAN_L_WH FP28
R90 / 1k	Indicator LED current limiter.	1: +3V3; 2: Net-(D10-A) FP28
R91 / 47k	Indicator base-current limiter.	1: Net-(Q6-B); 2: BAN_C_BK FP28
R92 / 1k	Indicator LED current limiter.	1: +3V3; 2: Net-(D11-A) FP28
R93 / 47k	Indicator base-current limiter.	1: Net-(Q7-B); 2: BAN_C_WH FP28
R94 / 1k	Indicator LED current limiter.	1: +3V3; 2: Net-(D12-A) FP28
R95 / 47k	Indicator base-current limiter.	1: Net-(Q8-B); 2: BAN_R_BK FP28
R96 / 1k	Indicator LED current limiter.	1: +3V3; 2: Net-(D13-A) FP28
R97 / 47k	Indicator base-current limiter.	1: Net-(Q9-B); 2: BAN_R_WH FP28
R98 / 1k	Indicator LED current limiter.	1: +3V3; 2: Net-(D14-A) FP28
U40 / SN74LVC2G08DCUR	Dual PWM enable AND gate.	See full pin atlas. FP24

Connector harness

Ref / value	Purpose	Connections / footprint
BANNER1 / Banner Q20NDXL	Purchased sensor; connect through its sheet 4, 5 or 8 harness.	External; see harness/sensor chapter. No PCB footprint
BANNER2 / Banner Q20NDXL	Purchased sensor; connect through its sheet 4, 5 or 8 harness.	External; see harness/sensor chapter. No PCB footprint
BANNER3 / Banner Q20NDXL	Purchased sensor; connect through its sheet 4, 5 or 8 harness.	External; see harness/sensor chapter. No PCB footprint
BAUM1 / Baumer O300.DI-GM1J.72N	Purchased sensor; connect through its sheet 4, 5 or 8 harness.	External; see harness/sensor chapter. No PCB footprint
BAUM2 / Baumer O300.DI-GM1J.72N	Purchased sensor; connect through its sheet 4, 5 or 8 harness.	External; see harness/sensor chapter. No PCB footprint
BAUM3 / Baumer O300.DI-GM1J.72N	Purchased sensor; connect through its sheet 4, 5 or 8 harness.	External; see harness/sensor chapter. No PCB footprint
ML11 / JSumo ML1	Purchased sensor; connect through its sheet 4, 5 or 8 harness.	External; see harness/sensor chapter. No PCB footprint
ML12 / JSumo ML1	Purchased sensor; connect through its sheet 4, 5 or 8 harness.	External; see harness/sensor chapter. No PCB footprint
ML13 / JSumo ML1	Purchased sensor; connect through its sheet 4, 5 or 8 harness.	External; see harness/sensor chapter. No PCB footprint
ML14 / JSumo ML1	Purchased sensor; connect through its sheet 4, 5 or 8 harness.	External; see harness/sensor chapter. No PCB footprint

Footprint dictionary

All assigned PCB footprints, including the local module and MAX14819 land patterns. These names identify the actual KiCad library entries; do not replace them solely on package appearance.

ID	KiCad footprint
FP1	Button_Switch_SMD:SW_SPST_TL3342
FP2	Capacitor_SMD:C_0805_2012Metric
FP3	Capacitor_SMD:C_1210_3225Metric
FP4	Capacitor_THT:CP_Radial_D6.3mm_P2.50mm
FP5	Capacitor_THT:CP_Radial_D8.0mm_P3.50mm
FP6	Connector_JST:JST_XH_B2B-XH-A_1x02_P2.50mm_Vertical
FP7	Connector_JST:JST_XH_B3B-XH-A_1x03_P2.50mm_Vertical
FP8	Connector_JST:JST_XH_B4B-XH-A_1x04_P2.50mm_Vertical
FP9	Connector_JST:JST_XH_B5B-XH-A_1x05_P2.50mm_Vertical
FP10	Connector_JST:JST_XH_B6B-XH-A_1x06_P2.50mm_Vertical
FP11	Connector_JST:JST_XH_B9B-XH-A_1x09_P2.50mm_Vertical
FP12	Converter_DCDC:Converter_DCDC_TRACO_TSR2-xxxx_THT
FP13	Crystal:Crystal_SMD_HC49-SD
FP14	Diode_SMD:D_SMA
FP15	Diode_SMD:D_SMB
FP16	Diode_SMD:D_SMC
FP17	Fuse:Fuse_1812_4532Metric
FP18	Fuse:Fuse_Littelfuse-NANO2-451_453
FP19	LED_SMD:LED_0805_2012Metric
FP20	Levi:MAX14819_T4877_4C_EP5.5
FP21	Levi:Pololu_S13V_5pin
FP22	Levi:Pololu_U3V50
FP23	Package_SO:SOIC-14_3.9x8.7mm_P1.27mm
FP24	Package_SO:VSSOP-8_2.3x2mm_P0.5mm
FP25	Package_TO_SOT_SMD:SOT-23
FP26	RF_Module:ESP32-C6-MINI-1
FP27	RF_Module:ESP32-S3-WROOM-1
FP28	Resistor_SMD:R_0805_2012Metric
FP29	TestPoint:TestPoint_Pad_D2.0mm

Full IC and connector pin atlas

Generated from embedded schematic symbols and the exported PCB netlist. Package pin numbers are the left column. NC indicates an unused, unconnected pin in this design. GPIO names are separate from pad numbers. Internal local nets retain their KiCad-generated names.

J1 - B2B-XH-A | power

Pin	Signal / pin name	Electrical net
1	Pin_1	BAT_FUSED
2	Pin_2	GND

J2 - B6B-XH-A | program

Pin	Signal / pin name	Electrical net
1	Pin_1	GND
2	Pin_2	+3V3
3	Pin_3	CTRL_PROG_TX
4	Pin_4	CTRL_PROG_RX
5	Pin_5	CTRL_EN
6	Pin_6	CTRL_BOOT

J3 - B6B-XH-A | program

Pin	Signal / pin name	Electrical net
1	Pin_1	GND
2	Pin_2	+3V3
3	Pin_3	TEL_PROG_TX
4	Pin_4	TEL_PROG_RX
5	Pin_5	TEL_EN
6	Pin_6	TEL_BOOT

J4 - B4B-XH-A | L

Pin	Signal / pin name	Electrical net
1	Pin_1	+24V_SENS
2	Pin_2	TEACH_L
3	Pin_3	GND
4	Pin_4	CQ_L

J5 - B4B-XH-A | C

Pin	Signal / pin name	Electrical net
1	Pin_1	+24V_SENS
2	Pin_2	TEACH_C
3	Pin_3	GND
4	Pin_4	CQ_C

J6 - B4B-XH-A | R

Pin	Signal / pin name	Electrical net
1	Pin_1	+24V_SENS
2	Pin_2	TEACH_R
3	Pin_3	GND
4	Pin_4	CQ_R

J7 - B9B-XH-A | selector

Pin	Signal / pin name	Electrical net
1	Pin_1	BAUM_L_DIGRAW
2	Pin_2	CQ_L
3	Pin_3	IOL_CQ_L
4	Pin_4	BAUM_C_DIGRAW
5	Pin_5	CQ_C
6	Pin_6	IOL_CQ_C
7	Pin_7	BAUM_R_DIGRAW
8	Pin_8	CQ_R
9	Pin_9	IOL_CQ_R

J8 - B4B-XH-A | L

Pin	Signal / pin name	Electrical net
1	Pin_1	+24V_SENS
2	Pin_2	BN_L_WH_RAW
3	Pin_3	GND
4	Pin_4	BN_L_BK_RAW

J9 - B4B-XH-A | C

Pin	Signal / pin name	Electrical net
1	Pin_1	+24V_SENS
2	Pin_2	BN_C_WH_RAW
3	Pin_3	GND
4	Pin_4	BN_C_BK_RAW

J10 - B4B-XH-A | R

Pin	Signal / pin name	Electrical net
1	Pin_1	+24V_SENS
2	Pin_2	BN_R_WH_RAW
3	Pin_3	GND
4	Pin_4	BN_R_BK_RAW

J11 - B3B-XH-A | FL

Pin	Signal / pin name	Electrical net
1	Pin_1	+3V3
2	Pin_2	GND
3	Pin_3	LINE_FL_RAW

J12 - B3B-XH-A | FR

Pin	Signal / pin name	Electrical net
1	Pin_1	+3V3
2	Pin_2	GND
3	Pin_3	LINE_FR_RAW

J13 - B3B-XH-A | RL

Pin	Signal / pin name	Electrical net
1	Pin_1	+3V3
2	Pin_2	GND
3	Pin_3	LINE_RL_RAW

J14 - B3B-XH-A | RR

Pin	Signal / pin name	Electrical net
1	Pin_1	+3V3
2	Pin_2	GND
3	Pin_3	LINE_RR_RAW

J15 - MDD20A control

Pin	Signal / pin name	Electrical net
1	Pin_1	GND
2	Pin_2	PWM_L
3	Pin_3	DIR_L
4	Pin_4	PWM_R
5	Pin_5	DIR_R

J16 - ARM removable link

Pin	Signal / pin name	Electrical net
1	Pin_1	MOTOR_ARM
2	Pin_2	MOTOR_ENABLE

U1 - U3V50F24

Pin	Signal / pin name	Electrical net
1	EN	NC
2	VIN	LOGIC_IN
3	GND	GND
4	GND	GND
5	VOUT	+24V_RAW

U2 - S13V30F5

Pin	Signal / pin name	Electrical net
1	EN	NC
2	VIN	LOGIC_IN
3	GND	GND
4	GND	GND
5	VOUT	+5V

U3 - TSR2-2433

Pin	Signal / pin name	Electrical net
1	Vin	LOGIC_IN
2	GND	GND
3	Vout	+3V3

U10 - ESP32-S3-WROOM-1-N8

Pin	Signal / pin name	Electrical net
1	GND	GND
2	3V3	+3V3
3	EN	CTRL_EN
4	IO4	LINE_RL_ADC
5	IO5	LINE_RR_ADC
6	IO6	BAT_ADC
7	IO7	BAUM_L
8	IO15	BAN_R_WH
9	IO16	IOL_CS_A
10	IO17	IOL_CS_B
11	IO18	IOL_SCK
12	IO8	BAUM_C
13	USB_D-	NC
14	USB_D+	NC
15	IO3	NC
16	IO46	NC
17	IO9	BAUM_R
18	IO10	BAN_L_BK
19	IO11	BAN_L_WH
20	IO12	BAN_C_BK
21	IO13	BAN_C_WH
22	IO14	BAN_R_BK
23	IO21	IOL_MOSI
24	IO47	CTRL_RX
25	IO48	MOTOR_ARM
26	IO45	NC

Pin	Signal / pin name	Electrical net
27	IO0	CTRL_BOOT
28	IO35	IOL_MISO
29	IO36	IOL_IRQ_A
30	IO37	IOL_IRQ_B
31	IO38	PWM_L_RAW
32	IO39	DIR_L_RAW
33	IO40	PWM_R_RAW
34	IO41	DIR_R_RAW
35	IO42	CTRL_TX
36	RXD0	CTRL_PROG_RX
37	TXD0	CTRL_PROG_TX
38	IO2	LINE_FR_ADC
39	IO1	LINE_FL_ADC
40	GND	GND
41	GND	GND

U11 - ESP32-C6-MINI-1-N4

Pin	Signal / pin name	Electrical net
1	GND	GND
2	GND	GND
3	3V3	+3V3
4	NC	NC
5	IO2	NC
6	IO3	NC
7	NC	NC
8	EN	TEL_EN
9	IO4	NC
10	IO5	NC
11	GND	GND
12	IO0	NC
13	IO1	NC
14	GND	GND
15	IO6	NC

Pin	Signal / pin name	Electrical net
16	IO7	NC
17	IO12	NC
18	IO13	NC
19	IO14	NC
20	IO15	NC
21	NC	NC
22	IO8	TEL_STRAP8
23	IO9	TEL_BOOT
24	IO18	TELEM_RX
25	IO19	TELEM_TX
26	IO20	NC
27	IO21	NC
28	IO22	NC
29	IO23	NC
30	RXD0	TEL_PROG_RX
31	TXD0	TEL_PROG_TX
32	NC	NC
33	NC	NC
34	NC	NC
35	NC	NC
36	GND	GND
37	GND	GND
38	GND	GND
39	GND	GND
40	GND	GND
41	GND	GND
42	GND	GND
43	GND	GND
44	GND	GND
45	GND	GND
46	GND	GND
47	GND	GND
48	GND	GND

Pin	Signal / pin name	Electrical net
49	GND	GND
50	GND	GND
51	GND	GND
52	GND	GND
53	GND	GND

U20 - LM339BIDR

Pin	Signal / pin name	Electrical net
1	~	BAUM_C
2	~	BAUM_L
3	V+	+24V_SENS
4	-	SENSE_REF
5	+	Net-(U20A-+)
6	-	SENSE_REF
7	+	Net-(U20B-+)
8	-	SENSE_REF
9	+	GND
10	-	SENSE_REF
11	+	Net-(U20C-+)
12	V-	GND
13	~	BAUM_R
14	~	NC

U21 - LM339BIDR

Pin	Signal / pin name	Electrical net
1	~	BAN_L_WH
2	~	BAN_L_BK
3	V+	+24V_SENS
4	-	SENSE_REF
5	+	Net-(U21A-+)
6	-	SENSE_REF
7	+	Net-(U21B-+)
8	-	SENSE_REF

Pin	Signal / pin name	Electrical net
9	+	Net-(U21D-+)
10	-	SENSE_REF
11	+	Net-(U21C-+)
12	V-	GND
13	~	BAN_C_BK
14	~	BAN_C_WH

U22 - LM339BIDR

Pin	Signal / pin name	Electrical net
1	~	BAN_R_WH
2	~	BAN_R_BK
3	V+	+24V_SENS
4	-	SENSE_REF
5	+	Net-(U22A-+)
6	-	SENSE_REF
7	+	Net-(U22B-+)
8	-	SENSE_REF
9	+	GND
10	-	SENSE_REF
11	+	GND
12	V-	GND
13	~	NC
14	~	NC

U30 - MAX14819AATM

Pin	Signal / pin name	Electrical net
1	~{IRQ}	IOL_IRQ_A
2	RXRDYA/LD1A	NC
3	RXERRA/LD2A	NC
4	RXRDYB/LD1B	NC
5	RXERRB/LD2B	NC
6	V5	+5V
7	REGEN	GND

Pin	Signal / pin name	Electrical net
8	VCC	+24V_SENS
9	G1A	NC
10	SN1A	PHY0_SN_A
11	SN2A	PHY0_SN_A
12	G2A	NC
13	L+A	GND
14	DIA	GND
15	NC	NC
16	NC	NC
17	CQA	IOL_CQ_L
18	NC	NC
19	NC	NC
20	CQB	IOL_CQ_C
21	NC	NC
22	NC	NC
23	DIB	GND
24	L+B	GND
25	G2B	NC
26	SN2B	PHY0_SN_B
27	SN1B	PHY0_SN_B
28	G1B	NC
29	VCC	+24V_SENS
30	LIB	NC
31	RXB	NC
32	TXENB	GND
33	TXB	GND
34	TXA	GND
35	TXENA	GND
36	RXA	NC
37	LIA	NC
38	A0	GND
39	A1	GND
40	VL	+3V3

Pin	Signal / pin name	Electrical net
41	XI	PHY0_XI
42	XO	PHY0_XO
43	CLKI	GND
44	CLKO	NC
45	SDI	IOL_MOSI
46	SDO	IOL_MISO
47	SCLK	IOL_SCK
48	~{CS}	IOL_CS_A
49	EP/GND	GND

U31 - MAX14819AATM

Pin	Signal / pin name	Electrical net
1	~{IRQ}	IOL_IRQ_B
2	RXRDYA/LD1A	NC
3	RXERRA/LD2A	NC
4	RXRDYB/LD1B	NC
5	RXERRB/LD2B	NC
6	V5	+5V
7	REGEN	GND
8	VCC	+24V_SENS
9	G1A	NC
10	SN1A	PHY1_SN_A
11	SN2A	PHY1_SN_A
12	G2A	NC
13	L+A	GND
14	DIA	GND
15	NC	NC
16	NC	NC
17	CQA	IOL_CQ_R
18	NC	NC
19	NC	NC
20	CQB	NC
21	NC	NC

Pin	Signal / pin name	Electrical net
22	NC	NC
23	DIB	GND
24	L+B	GND
25	G2B	NC
26	SN2B	PHY1_SN_B
27	SN1B	PHY1_SN_B
28	G1B	NC
29	VCC	+24V_SENS
30	LIB	NC
31	RXB	NC
32	TXENB	GND
33	TXB	GND
34	TXA	GND
35	TXENA	GND
36	RXA	NC
37	LIA	NC
38	A0	GND
39	A1	GND
40	VL	+3V3
41	XI	PHY1_XI
42	XO	PHY1_XO
43	CLKI	GND
44	CLKO	NC
45	SDI	IOL_MOSI
46	SDO	IOL_MISO
47	SCLK	IOL_SCK
48	~{CS}	IOL_CS_B
49	EP/GND	GND

U40 - SN74LVC2G08DCUR

Pin	Signal / pin name	Electrical net
1	~	PWM_L_RAW
2	~	MOTOR_ENABLE

Pin	Signal / pin name	Electrical net
3	~	Net-(R72-Pad2)
4	GND	GND
5	~	PWM_R_RAW
6	~	MOTOR_ENABLE
7	~	Net-(R69-Pad2)
8	VCC	+3V3

Schematic sheet collection

The next ten pages reproduce the saved schematic at its original A3 page size using vector artwork. Zoom rather than printing multiple sheets on one small page. The PDF sidebar has a bookmark for each sheet. Return to the relevant teaching chapter when tracing a repeated circuit.

Sheet	Source file
1	Levi.kicad_sch
2	01_Control.kicad_sch
3	02_Telemetry.kicad_sch
4	03_Baumer.kicad_sch
5	04_Banner.kicad_sch
6	05_IOLink_A.kicad_sch
7	06_IOLink_B.kicad_sch
8	07_Line_Motor.kicad_sch
9	08_Harness.kicad_sch
10	09_Enable_Debug.kicad_sch